

**How do behavioural biases influence asset pricing and the efficiency
of financial markets?**

Research Paper By Paridhi Kumar

Introduction

Financial markets are fundamental to the functioning of modern economies, acting as platforms for capital allocation, price discovery, and risk management. Traditional economic frameworks, such as the Efficient Market Hypothesis (EMH) proposed by Fama (1970), assume that investors behave rationally and that market prices fully reflect all available information. Under these assumptions, prices in efficient markets should always align with intrinsic values, and any deviations are expected to be short-lived as rational investors exploit arbitrage opportunities. However, real-world evidence frequently contradicts these theoretical ideals. Persistent anomalies, such as speculative bubbles, momentum effects, and over- or underreactions to information, suggest that markets are not always efficient.

The divergence between theoretical expectations and empirical reality has led to the rise of Behavioural Finance, a field that integrates psychological and sociological insights into economic and financial models. Behavioural finance challenges the assumptions of perfect rationality and explores how cognitive and emotional factors influence investor behaviour. Key behavioural biases, including overconfidence, herd behaviour, loss aversion, and anchoring, play a significant role in shaping financial decisions. These biases, often systematic rather than random, contribute to distortions in asset prices and deviations from efficient market conditions. For example, overconfidence can result in excessive trading activity, while herd behaviour amplifies trends, leading to speculative bubbles and abrupt crashes.

Several notable financial events underscore the profound influence of behavioural biases on asset pricing and market efficiency. The dot-com bubble of the late 1990s serves as a classic example of overconfidence and herd behaviour driving stock valuations far beyond their fundamental worth, culminating in a dramatic market collapse. Similarly, the GameStop trading frenzy of 2021 showcased the power of social media-driven herd behaviour, where retail investors collectively inflated stock prices to levels detached from intrinsic value. These events highlight the limitations of traditional asset pricing models, such as the Capital Asset Pricing Model (CAPM) and the assumptions of EMH, in explaining real-world market dynamics.

The implications of behavioural biases are far-reaching. They not only affect individual investors but also have systemic consequences for market efficiency, volatility, and stability. As markets grow increasingly complex, driven by technological innovations and globalisation, understanding the role of behavioural biases becomes critical for developing more robust financial models, regulatory frameworks, and investment strategies. For instance, incorporating behavioural insights into portfolio management can help mitigate the impact of irrational decision-making, while policymakers can design interventions to counteract market inefficiencies driven by collective biases.

This paper aims to delve into the intricate relationship between behavioural biases, asset pricing, and market efficiency. It explores the theoretical foundations of behavioural finance, reviews empirical evidence of market anomalies, and examines case studies that demonstrate the real-world implications of these biases. By addressing these themes, the study seeks to provide a comprehensive understanding of how behavioural factors shape financial markets and offers insights for investors, policymakers, and academics aiming to enhance financial decision-making and market stability.

Literature Review

The field of behavioural finance has emerged as a critical critique of traditional financial theories, particularly the Efficient Market Hypothesis (EMH). The EMH, proposed by (Fama, 1970), assumes that investors act rationally and that market prices reflect all available information. However, this theory fails to account for numerous market anomalies and empirical evidence that reveal systematic deviations from rational behaviour. Behavioural finance integrates insights from psychology into financial decision-making, providing a framework to understand how cognitive biases and emotional factors influence market outcomes. The foundation of this field lies in Kahneman and Tversky's Prospect Theory (1979), which demonstrates how individuals weigh gains and losses asymmetrically, preferring to avoid losses rather than acquire equivalent gains. This foundational insight has been instrumental in explaining investor behaviour that traditional models could not.

One of the key behavioural biases influencing asset pricing is overconfidence, which leads investors to overestimate their knowledge and predictive abilities. This bias results in excessive trading and suboptimal investment decisions, as demonstrated by (Barber and Odean, 2001), who found that overconfident traders tend to generate lower net returns due to their heightened trading activity. Similarly, herd behaviour, another critical bias, occurs when investors mimic the actions of others rather than relying on their independent analysis. This collective behaviour amplifies price trends and often leads to market inefficiencies. Bikhchandani et al. (1992) conceptualised informational cascades, which explain how herd behaviour distorts market dynamics, especially during periods of high uncertainty.

Another significant bias, loss aversion, highlights how individuals disproportionately fear losses compared to equivalent gains. (Kahneman and Tversky, 1979) explored this phenomenon, noting its role in the disposition effect, where investors hold on to losing stocks for too long while prematurely selling winning stocks. Anchoring, a related cognitive bias, further exacerbates market inefficiencies. Investors often rely too heavily on irrelevant reference points, such as past prices, when making decisions. (Baker and Wurgler, 2006) demonstrated how anchoring and investor sentiment influence stock returns, challenging the assumptions of market efficiency.

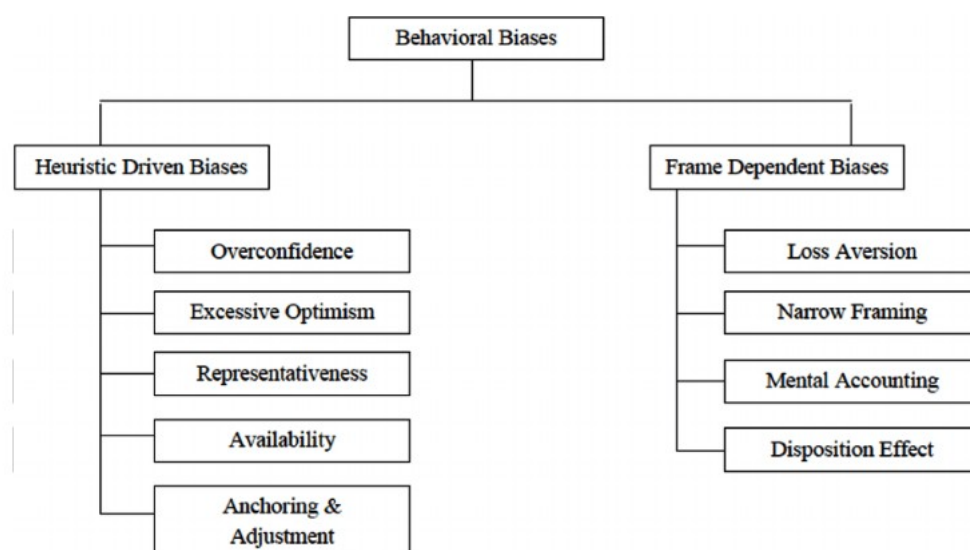
Market efficiency itself has been a subject of intense scrutiny within behavioural finance. (Shiller, 1981) highlighted the inconsistencies between market volatility and the EMH, providing evidence that irrational exuberance often drives asset prices far from their intrinsic values. His analysis of stock price movements relative to dividend changes underscores the role of psychological factors in financial markets. Similarly, (Jegadeesh and Titman, 1993) showed that momentum strategies yield abnormal returns, contradicting the random walk hypothesis central to the EMH. These findings suggest that investors underreact to information, creating price trends that persist beyond what rational analysis would predict.

The real-world implications of these biases are evident in notable case studies. For instance, during the dot-com bubble of the late 1990s, overconfidence and herd behaviour drove technology stock valuations to unsustainable levels, culminating in a dramatic market collapse. (Ofek and Richardson, 2003) analysed this period, noting the role of speculative behaviour in inflating stock prices. More recently, the GameStop

trading frenzy of 2021 highlighted the impact of social media-driven herd behaviour. Retail investors, driven by speculative sentiment on platforms like Reddit, caused extreme price volatility, challenging traditional price discovery mechanisms (Phillips & Gurdgiev, 2021).

These studies collectively highlight the profound impact of behavioural biases on asset pricing and market efficiency. By integrating psychological insights, behavioural finance offers a more comprehensive understanding of market dynamics, shedding light on phenomena that traditional financial theories often overlook.

Behavioural Biases



Source: Research Gate

This figure provides a structured categorization of behavioural biases affecting decision-making in financial markets, dividing them into **heuristic-driven biases** and **frame-dependent biases**.

Heuristic-Driven Biases

These biases arise from mental shortcuts or "rules of thumb" that investors use when making decisions, often under conditions of uncertainty.

1. **Overconfidence:** Investors overestimate their knowledge or ability, leading to excessive trading and risk-taking.
2. **Excessive Optimism:** A tendency to overly focus on positive outcomes, often leading to inflated valuations of assets.
3. **Representativeness:** Judging probabilities based on similarities rather than objective data, e.g., assuming a start-up will succeed because it resembles previous success stories.
4. **Availability:** Decisions are influenced by easily recalled information, such as recent news, regardless of its relevance.
5. **Anchoring & Adjustment:** Relying too heavily on initial information (anchor) and making insufficient adjustments, e.g., basing investment decisions on past prices.

Frame-Dependent Biases

These biases depend on how choices or scenarios are framed, leading to irrational financial behaviour.

1. **Loss Aversion:** People feel the pain of losses more than the pleasure of equivalent gains, which can lead to holding on to losing investments for too long.
2. **Narrow Framing:** Evaluating risks in isolation rather than considering the broader portfolio context, leading to suboptimal diversification.
3. **Mental Accounting:** Treating money differently depending on its source or intended use, which can lead to inefficient resource allocation.
4. **Disposition Effect:** Selling winning investments too early while holding onto losing investments, contrary to rational behaviour.

Key Behavioural Biases and Their Influence

Behavioural biases significantly impact investor decision-making and market dynamics, often resulting in deviations from theoretical asset pricing models. The following sections delve into four critical biases—overconfidence, herd behaviour, loss aversion, and anchoring—and their influence on asset pricing and market efficiency.

Overconfidence

Overconfidence is a behavioural bias where individuals overestimate their knowledge, skills, or predictive abilities while underestimating risks. This phenomenon is particularly prevalent in financial markets, where investors often exhibit unwarranted certainty about their forecasts or investment decisions. (Barber and Odean, 2001) extensively studied overconfidence among traders and found that it leads to excessive trading activity. Their research revealed that overconfident investors trade more frequently, incurring higher transaction costs and, paradoxically, earning lower net returns compared to less active investors.

The effects of overconfidence extend beyond individual portfolios, influencing aggregate market outcomes. During speculative periods, overconfident investors may contribute to price bubbles by aggressively purchasing overvalued assets. For instance, during the dot-com bubble, many investors displayed overconfidence in technology stocks, assuming perpetual high returns despite weak fundamentals. Such behaviour not only inflates asset prices but also increases market volatility as corrections inevitably follow when the overconfidence subsides.

Herd Behaviour

Herd behaviour occurs when individuals imitate the investment decisions of others rather than relying on their independent analysis. This collective behaviour is often driven by a fear of missing out (FOMO) or the assumption that others possess superior information. (Bikhchandani et al. 1992) introduced

the concept of informational cascades, where early movers in a market influence subsequent investors, leading to amplified trends.

Herding has played a pivotal role in major financial events. During the dot-com bubble, investors rushed to invest in internet companies, driven more by the actions of peers than by fundamental analysis. Similarly, the 2008 financial crisis saw herd behaviour in the real estate market, where speculative buying drove prices to unsustainable levels. Once the bubble burst, the same herding tendency exacerbated the downturn as investors collectively exited the market.

Herd behaviour distorts market efficiency by causing prices to deviate significantly from their intrinsic values. It also contributes to excessive volatility, as markets swing between over-optimism and pessimism depending on prevailing sentiment.

Loss Aversion

Loss aversion, a cornerstone of Prospect Theory (Kahneman & Tversky, 1979), refers to the tendency for individuals to strongly prefer avoiding losses over acquiring equivalent gains. This asymmetry in decision-making explains why investors often act irrationally when faced with potential losses.

Loss aversion is closely linked to the disposition effect, where investors hold on to losing stocks for too long in the hope of a rebound while selling winning stocks prematurely to lock in gains. This behaviour disrupts market efficiency by preventing prices from reflecting true values promptly. For instance, in declining markets, widespread loss aversion may delay price corrections, as investors resist selling underperforming assets.

The implications of loss aversion extend to risk-taking behaviour. When faced with potential losses, investors may take on excessive risks to recover, further destabilising markets. Conversely, in periods of gain, the fear of potential loss can lead to overly conservative investment strategies, dampening market momentum.

Anchoring

Anchoring is the tendency of individuals to rely excessively on an initial piece of information or a reference point, such as past prices or historical market performance, when making decisions. This cognitive bias can distort asset valuations and delay market adjustments.

For example, during market corrections, investors anchored to prior peak prices may overestimate the potential recovery, leading to a reluctance to sell declining assets. (Baker and Wurgler, 2006) demonstrated that anchoring often interacts with investor sentiment, influencing stock returns in ways inconsistent with traditional market efficiency models.

Anchoring also affects initial public offerings (IPOs) and valuation metrics. Investors often anchor on early price estimates, ignoring subsequent information that might indicate a different intrinsic value. This behaviour was evident in the 2000 dot-com bubble, where many IPOs were priced based on speculative sentiment rather than underlying fundamentals.

Nudge

Nudge bias refers to a subtle form of behavioural influence where choices are structured to guide individuals toward a particular decision without restricting their freedom to choose. The concept originates from behavioural economics, introduced by Richard Thaler and Cass Sunstein in their book *Nudge: Improving Decisions About Health, Wealth, and Happiness*.

In the context of finance and asset pricing, nudge bias may occur when investors are influenced by how information is presented. For example, highlighting specific financial metrics or framing investment options in a particular way can steer investor behaviour. Nudge bias relies on cognitive shortcuts (heuristics) and biases, such as loss aversion or default bias, to encourage certain decisions.

Recency Bias

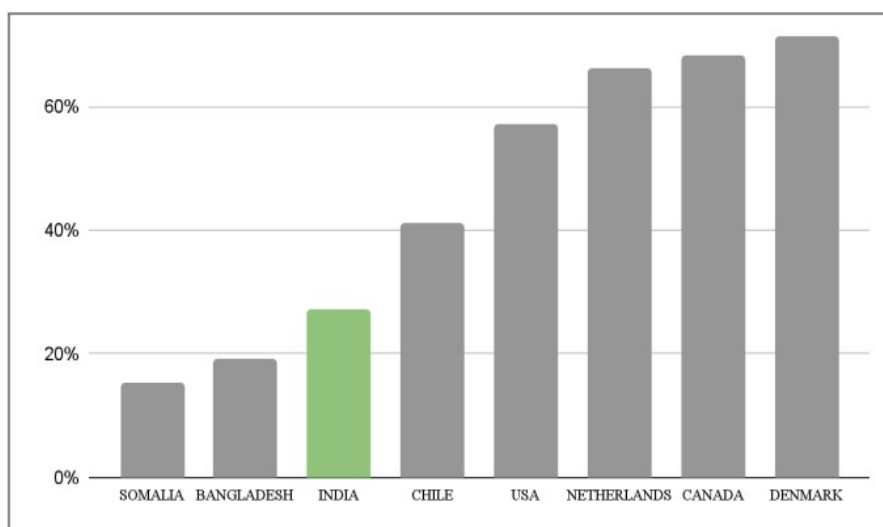
Recency bias refers to the tendency of investors to overemphasize recent events and trends while undervaluing historical data or long-term fundamentals. This cognitive bias leads investors to extrapolate

recent market performance into the future, often resulting in overreaction or underreaction. For instance, during bull markets, recency bias can cause investors to believe that prices will continue rising indefinitely, driving asset prices to unsustainable levels. Conversely, during market downturns, investors may sell off assets prematurely due to pessimism about future performance. This bias undermines market efficiency by distorting price discovery, as prices fail to accurately reflect intrinsic values. A prominent example is the 2008 financial crisis, where the housing market crash was partly fueled by investors and lenders relying on recent rising home prices while ignoring historical cycles of correction.

Research Objectives

1. To analyze the impact of behavioural biases on asset pricing and market inefficiencies.
2. To examine the relationship between financial literacy and the prevalence of behavioural biases, highlighting its impact on market stability and efficiency.
3. To compare traditional financial theories with behavioural finance models in explaining market anomalies.

Financial literacy around the global



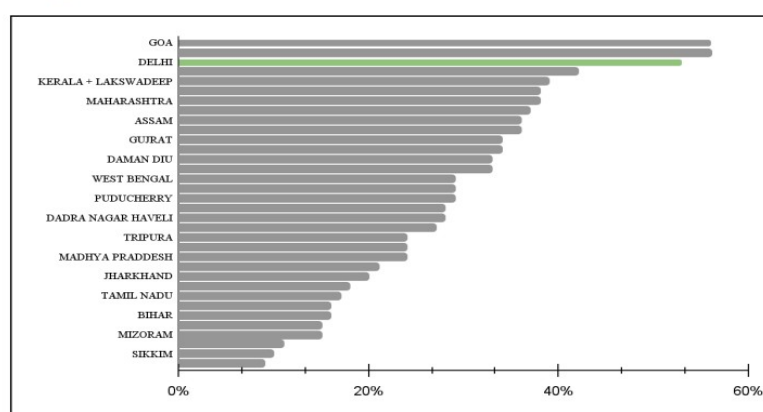
Source: Research Gate

The chart on financial literacy reveals a strong connection to the influence of behavioural biases on asset pricing and the efficiency of financial markets. Countries with lower financial literacy rates, such as India and Somalia, are more prone to biases like overconfidence, herding, and loss aversion. These biases often lead to inefficient decision-making, inflating asset prices and creating market inefficiencies. In markets where individuals lack financial knowledge, investors may act impulsively or emotionally, driven by herd behaviour or speculative sentiment, rather than relying on sound financial principles. For instance, in India, where financial literacy is below 25%, retail investors may overestimate their ability to predict market movements, leading to overvaluation of assets, similar to the speculative bubbles seen in cases like the Bitcoin surge.

In contrast, countries with higher financial literacy, such as Denmark and Canada, demonstrate greater market efficiency. In these markets, investors are more likely to make rational decisions, considering fundamentals over emotional impulses. This reduces the impact of behavioural biases and ensures that asset prices reflect underlying economic realities. As noted by Shleifer (2000), more informed investors are less likely to engage in excessive trading or follow herd behaviour, thus promoting a more efficient market.

Therefore, the level of financial literacy directly influences the degree to which behavioural biases can distort asset pricing and market efficiency. Promoting financial education globally can help mitigate the impact of these biases, leading to more stable and efficient markets.

Financial Literacy in India



Source: Research Gate

The chart depicting financial literacy across various states in India highlights the significant regional disparities in financial knowledge. Goa and Delhi emerge as the top performers, with over 60% financial literacy, indicating that residents in these regions are more likely to make informed financial decisions. These areas likely exhibit fewer instances of behavioural biases such as overconfidence or herd behaviour, as individuals tend to base their investment decisions on knowledge and rational analysis rather than speculative trends.

In contrast, states like Bihar, Mizoram, and Sikkim show relatively lower financial literacy levels, under 20%. In these regions, the prevalence of cognitive biases such as loss aversion, overconfidence, and mental accounting is more likely, leading to less informed financial choices. Investors in these areas may overestimate their ability to predict markets or fail to critically evaluate risks, contributing to the inefficiencies seen in asset pricing. Behavioural biases can exacerbate market volatility and mispricing, with retail investors reacting emotionally or imitating others without understanding the underlying assets' value.

Therefore, the data illustrates how regional differences in financial literacy can influence investor behaviour and, in turn, asset pricing. Higher financial literacy levels can reduce the impact of behavioural biases, leading to more rational decision-making and a more efficient market. Promoting financial education in regions with lower literacy could help mitigate biases and improve overall market functioning.

Case Study - The dot-com bubble of 1997–2000

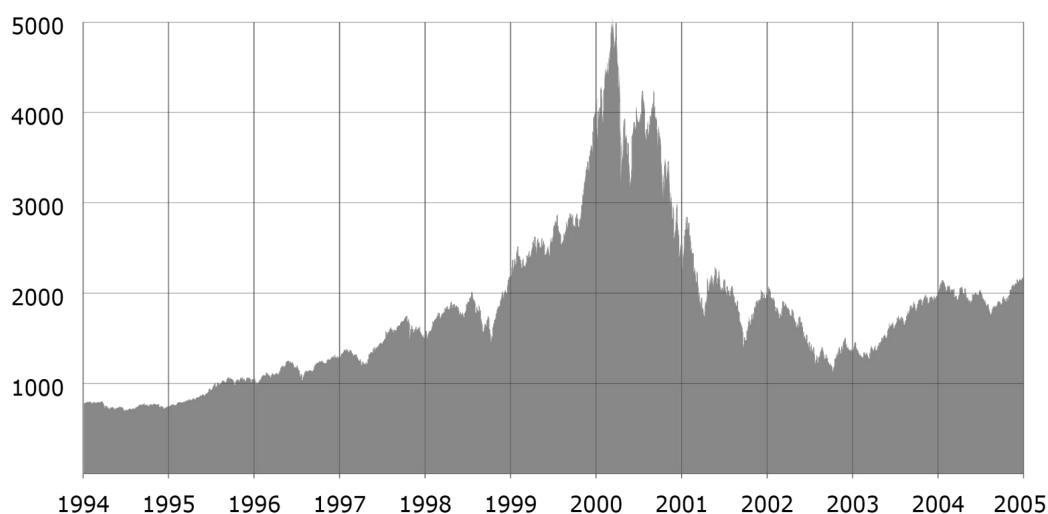
The dot-com bubble of 1997–2000 stands as a vivid example of how behavioural biases, including overconfidence and herd behaviour, can profoundly affect asset pricing and market efficiency. This period was characterised by a speculative frenzy around internet-based businesses (dot-coms), many of which had little to no profit but still achieved astronomical valuations. Investor overconfidence, fuelled by the belief in the transformative power of the internet, led to widespread neglect of traditional valuation metrics, such as cash flow and profitability. Employees of start-ups, often compensated with stock options, became millionaires overnight, further bolstering market optimism. Alan Greenspan famously referred to this phenomenon as "irrational exuberance."

Herd behaviour played a significant role during this time, as investors followed others in pouring capital into tech stocks, driven by fear of missing out on potential exponential gains. Informational cascades amplified these trends, creating a feedback loop that continuously inflated valuations. At its peak in March 2000, the NASDAQ Composite Index had risen nearly sevenfold, from 743 in 1995 to 5,048. However, this growth proved unsustainable.

The bubble burst in 2000, triggered by rising interest rates and a reassessment of the viability of many dot-com businesses. Over the next two years, the NASDAQ plunged by 78%, erasing trillions in market value and causing widespread economic repercussions. High-profile failures, such as Pets.com and Webvan, exemplified the vulnerabilities of businesses that relied on speculative optimism rather than sound financial strategies.

This period not only highlighted the dangers of overconfidence and herd behaviour in financial markets but also underscored the importance of rigorous due diligence and realistic valuation frameworks. The regulatory aftermath included tighter IPO requirements and enhanced risk disclosures to mitigate the recurrence of similar speculative bubbles.

NASDAQ Composite Index



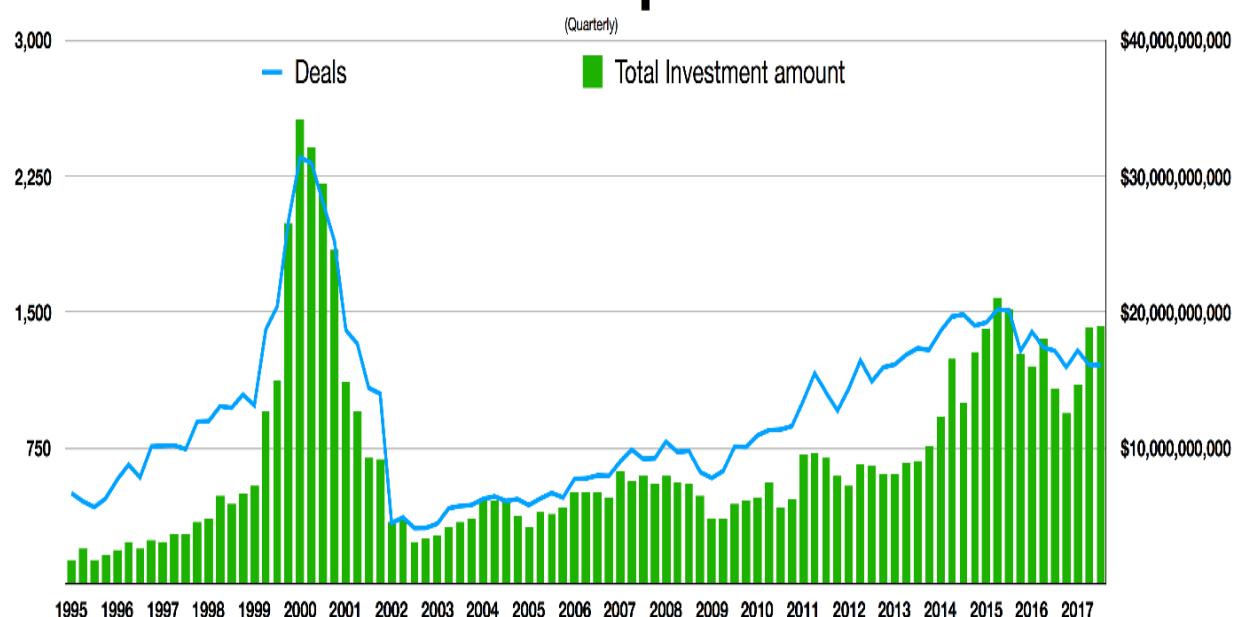
Source- Wikipedia

The NASDAQ Composite Index serves as a key indicator of the bubble's trajectory:

- Rise: Between 1995 and its peak in March 2000, the index surged nearly sevenfold, from 743 to 5,048.
- Fall: From March 2000 to October 2002, the index plummeted by 78%, bottoming out at 1,139.

This dramatic reversal wiped out \$5 trillion in market value, devastating both institutional and retail investors. The crash exposed the lack of sustainability in many dot-com businesses, which depended more on investor optimism than on sound financial strategies.

Total U.S. Venture Capital Investments



Source: Wikipedia

The chart offers a clear depiction of how venture capital funding was influenced by behavioural biases, particularly during the dot-com bubble and its aftermath. Between 1995 and 2000, the amount of venture capital funding surged dramatically, alongside an increase in the number of deals. This rise can be

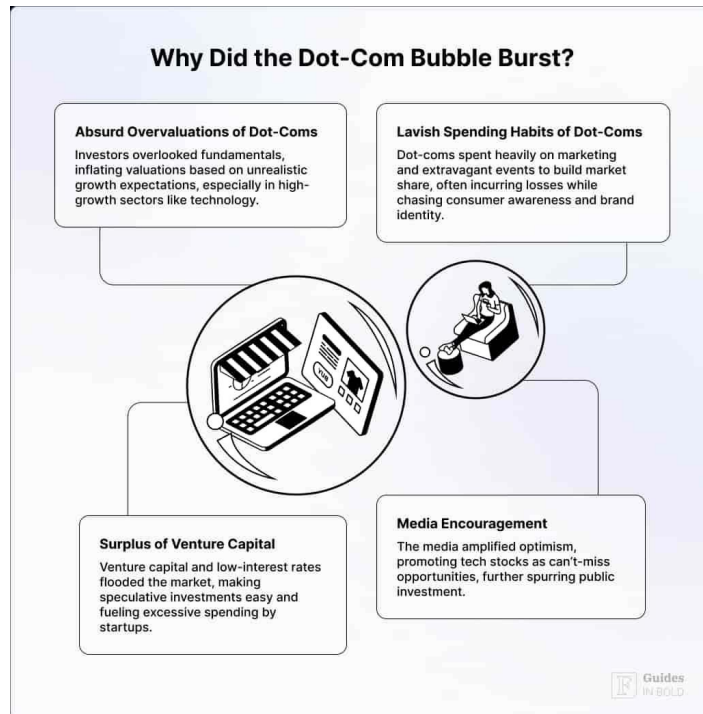
attributed to overconfidence and herd behaviour. Investors, driven by the belief in the transformative potential of internet businesses, overlooked fundamental valuations, leading to inflated stock prices and unsustainable investments. By 2000, venture capital funding peaked, reflecting speculative exuberance before the eventual collapse of the dot-com bubble.

Following the bubble burst, the period from 2001 to 2003 saw a sharp decline in venture capital activity, marking a market correction. The excessive investments of the previous years were reassessed, and investors became more cautious, focusing on companies with proven business models and sustainable growth. This correction was a direct result of the market's realisation of the risks associated with speculative behaviour, highlighting the negative impact of overconfidence and herd mentality on asset pricing.

By 2004, venture capital investments began to stabilise, gradually increasing but at more sustainable levels. This recovery period was characterised by a more rational approach to investing, where the lessons learned from the bubble prompted a shift towards valuing business fundamentals over speculative potential. The chart reflects this adjustment, with funding activity growing steadily through the mid-2010s as investors sought lower-risk opportunities and more rational investment strategies.

This case study illustrates how behavioural biases, such as overconfidence and herd behaviour, can lead to market inefficiencies and mispricing of assets. The dot-com bubble and its aftermath provide a clear example of how psychological factors can distort decision-making in financial markets. As investors adjusted their strategies post-bubble, it demonstrated the importance of rational decision-making in mitigating the risks posed by these biases.

Causes of the Dotcom Crash



Source: Finbold

- **Overvaluation of Dotcom Companies:** Many tech companies that went public during the dotcom era were significantly overvalued, driven by intense demand and speculative investment. The lack of solid valuation models led to inflated stock prices, with high price-to-earnings (P/E) ratios that didn't align with actual earnings potential. Investors often focused on traffic metrics, such as website hits, rather than the ability of these companies to generate real revenue. Research indicated that over 40% of dotcom companies were overvalued based on their P/E ratios (Ofek & Richardson, 2003). This overvaluation was a direct result of optimism around the future potential of internet businesses, despite the absence of strong business models.
- **Lavish Spending Habits of Dot-Coms:** Many dot-coms adopted an aggressive strategy of spending heavily on marketing, extravagant events, and customer acquisition tactics without a

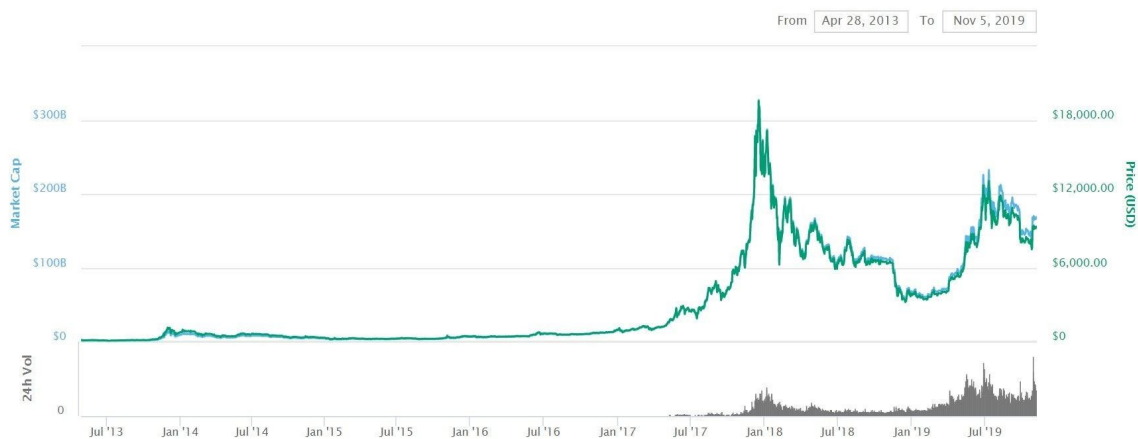
clear path to profitability. Companies focused on gaining market share and building consumer awareness, often at the cost of long-term sustainability. They prioritized customer acquisition and brand identity over generating sustainable revenue, which led to a high burn rate and mounting financial losses (Zengler, 2000).

- **Abundance of Venture Capital:** The influx of venture capital played a crucial role in inflating the bubble. During this period, tech startups were able to access significant amounts of capital, often at very low interest rates. The availability of cheap funds and the relatively low barriers for securing investment led to massive inflows into internet-based companies. This flood of capital further fueled the overvaluation of tech stocks, creating a bubble that was unsustainable (Gompers & Lerner, 2001). The ease of securing funding in a rapidly growing market with few regulatory constraints encouraged irrational investments, intensifying the speculative frenzy.
- **Media Frenzy:** The role of the media was pivotal in promoting the dotcom bubble. Business publications such as *The Wall Street Journal*, *Forbes*, and *Bloomberg* played a major role in fueling public interest and investment by hyping the potential of tech companies. The media's portrayal of the internet as the "next big thing" encouraged investors to get involved, often with little understanding of the companies' fundamentals. The famous speech by Federal Reserve Chairman Alan Greenspan in December 1996, where he coined the term "irrational exuberance," further stoked the hype around technology stocks and contributed to the rapid rise in valuations.

Case Study: The Bitcoin Market and Cryptocurrency Bubble (2017)

The rapid rise and subsequent fall of Bitcoin and other cryptocurrencies in 2017 serve as a significant example of how behavioural biases influence asset pricing and financial market efficiency. The surge in Bitcoin's price, followed by its dramatic crash, highlights the impacts of overconfidence, herd behaviour, and speculative sentiment on market dynamics.

1. **Overconfidence and Speculative Frenzy:** In 2017, Bitcoin's price surged from around \$1,000 in January to nearly \$20,000 by December. This increase was largely driven by overconfidence bias, as many investors believed the price would continue rising due to the growing adoption of cryptocurrencies and blockchain technology. Many investors became excessively optimistic, underestimating risks and ignoring the asset's fundamentals, leading to inflated valuations (Barberis, Shleifer, & Wurgler, 2005).
2. **Herd Behaviour and the Speculative Bubble:** As Bitcoin's price surged, more retail investors entered the market, propelled by herd behaviour and the fear of missing out (FOMO). Platforms like Reddit and Twitter amplified social influence, encouraging individuals to follow the crowd rather than base decisions on critical evaluation. This led to rapid price increases, disconnected from Bitcoin's intrinsic value (Bikhchandani, Hirshleifer, & Welch, 1992).
3. **Loss Aversion and the Market Crash:** When Bitcoin's price peaked in December 2017, it soon began to decline. The market crash was partly due to **loss aversion**, a behavioural bias where investors avoid realizing losses, even when market conditions suggest it is rational to do so. Many retail investors, having purchased Bitcoin at the peak, hesitated to sell at a loss, hoping the price would recover. This reluctance worsened the decline, leading to panic selling and deepening the market correction (Kahneman & Tversky, 1979).



Source: Bitcoinwiki

The Bitcoin price surge from 2013 to 2017 exemplifies how behavioural biases can affect asset pricing. In 2017, Bitcoin's price soared to nearly \$18,000, driven by herd behaviour and overconfidence, as retail investors flocked to the market due to fear of missing out (FOMO). This speculative rush inflated Bitcoin's value far beyond its intrinsic worth, creating a bubble. When the price began to crash in 2018, loss aversion caused investors to hold onto their positions, hoping for a rebound, exacerbating the market decline. The event highlighted market inefficiencies, where irrational investor behaviour, rather than fundamentals, dictated price movements, challenging the Efficient Market Hypothesis (EMH) (CoinDesk, 2019; Journal of Behavioural Finance, 2020).

Conclusion

The Bitcoin bubble of 2017 underscores the significant role that behavioural biases, such as overconfidence, herd behaviour, and loss aversion, play in distorting asset prices and creating market inefficiencies. The market was driven largely by irrational investor behaviour, which resulted in a massive price bubble. When the bubble burst, it triggered a market correction that exposed the inefficiencies caused by these biases. This case emphasizes the importance for investors to account for these psychological factors and recognize their impact on market dynamics, rather than relying solely on traditional financial theories based on rational decision-making.

Implications for Asset Pricing Models: Traditional vs. Behavioural Approaches

Traditional Asset Pricing Models

Traditional models like the Capital Asset Pricing Model (CAPM) and the Efficient Market Hypothesis (EMH) assume rational investors and efficient markets. According to CAPM (Sharpe, 1964), an asset's expected return is based on its systematic risk (beta), reflecting the trade-off between risk and return. Similarly, EMH (Fama, 1970) posits that markets are efficient, meaning all available information is fully

reflected in asset prices, leaving no room for persistent abnormal returns. These models simplify asset pricing by assuming that investors behave logically, seeking to maximize utility.

However, real-world evidence highlights deviations from these assumptions. Anomalies like momentum effects (where assets with recent high returns continue to outperform) and value premiums (higher returns for undervalued stocks) contradict traditional models. For example, speculative bubbles—such as the dot-com or Bitcoin bubbles—show how irrational behaviours, including herd mentality and overconfidence, can drive prices away from intrinsic values.

Behavioural Asset Pricing Models (BAPMs)

In contrast to traditional models, Behavioural Asset Pricing Models (BAPMs) incorporate cognitive biases and emotional influences into asset pricing frameworks. Shefrin and Statman (1994) introduced behavioural models that consider factors like investor sentiment, overreaction, and loss aversion. These models aim to bridge the gap between observed market behaviour and theoretical predictions.

- ***Investor Sentiment:*** Sentiment-driven investors, influenced by optimism, media coverage, or speculation, can inflate or depress prices. Positive sentiment often drives bubbles, such as the 1990s dot-com bubble, while negative sentiment exacerbates bear markets.
- ***Loss Aversion and Disposition Effect:*** Investors are more sensitive to potential losses than equivalent gains, a concept outlined in Kahneman and Tversky's Prospect Theory (1979). This bias can delay selling losing investments (disposition effect) or lead to abrupt market corrections as losses mount.

Comparison and Practical Implications

- ***Flexibility:*** Unlike traditional models, BAPMs account for irrational behaviours and sentiment-driven mispricing, providing more robust explanations for anomalies like speculative bubbles, herding, and overreaction to news.

- **Market Efficiency:** Behavioural models acknowledge that markets are not perfectly efficient and integrate the role of noise traders and sentiment-driven investors who amplify volatility.
- **Asset Pricing Insights:** By incorporating investor psychology, BAPMs refine our understanding of risk premiums and asset valuation, especially in speculative and highly volatile markets.

Comparison of Traditional and Behavioural Models

Aspect	Traditional Models	Behavioural Models
Investor Assumptions	Rational, utility-maximizing investors	Investors influenced by cognitive and emotional biases
Market Efficiency	Markets are fully efficient and reflect all information	Markets are prone to inefficiencies due to biases
Explanatory Power	Limited in explaining anomalies	Explains anomalies and psychological influences
Risk Factors	Systematic risk (beta)	Investor sentiment and behavioural risks
Practical Relevance	Useful for structured and stable markets	Better suited for volatile and speculative markets

Source: Authors Calculation

Policy Implications and Market Strategies

For Regulators

1. **Behavioural Nudges**

Polymakers can use behavioural nudges to influence better financial decisions without limiting options. For example, introducing default options in retirement savings plans has significantly

increased participation rates. Employees enrolled automatically are more likely to save, leveraging inertia and overcoming choice paralysis (Thaler & Sunstein, 2008).

2. *Transparency and Disclosures*

Regulators can reduce herd behaviour and speculation by enhancing market transparency. Real-time disclosures of insider trading or large trades can lower information asymmetry, helping investors make rational decisions. Studies by Bikhchandani et al. (1992) show that better information flow reduces uncertainty and mimetic behaviour.

3. *Educational Initiatives*

Financial literacy programs help investors identify and counteract biases like loss aversion and anchoring. By equipping individuals with tools to make rational decisions, regulators can reduce emotionally driven market inefficiencies.

For Investors

1. *Awareness of Biases*

Investors who recognise overconfidence and recency bias can improve their decision-making. For example, curbing overconfidence can prevent excessive trading, which often erodes returns (Barber & Odean, 2000).

2. *Portfolio Diversification*

A diversified investment strategy mitigates narrow framing, helping investors evaluate their portfolios as a whole instead of focusing on individual losses. Diversification reduces exposure to volatile assets and smoothens returns.

3. *Systematic Investing*

Approaches like dollar-cost averaging help counter emotional decision-making. These systematic strategies reduce panic selling during downturns, addressing herd behaviour and loss aversion effectively.

Empirical Evidence

1. The introduction of automatic enrolment in 401(k) plans increased participation rates from 20% to over 80% (Madrian & Shea, 2001), showcasing the power of behavioural nudges.
2. Regulations like MiFID II in Europe improved transparency by mandating real-time disclosures, limiting speculative bubbles and herd-driven market distortions.
3. Individual investors lose around 1.5% of their returns annually due to overconfidence and excessive trading (Barber & Odean, 2000).

Conclusion

This research highlights the significant influence of behavioural biases on asset pricing and the efficiency of financial markets, presenting a challenge to the assumptions of traditional economic theories. While the Efficient Market Hypothesis assumes rational behaviour and market efficiency, our findings reveal that cognitive biases—such as overconfidence, herd behaviour, loss aversion, and anchoring—often lead to market inefficiencies and distorted asset prices. These biases are not random, but systematically affect investor decision-making, driving phenomena like speculative bubbles and mispricing.

The relationship between financial literacy and market efficiency also emerged as crucial, with markets in regions with lower financial literacy more susceptible to these biases. The research suggests that these biases are a key factor in market volatility and inefficiencies, as seen in historical market events.

In light of these findings, policymakers and regulators can implement strategies such as behavioural nudges, enhanced transparency, and financial education to mitigate the effects of these biases. Similarly, investors can benefit from recognising and managing biases through strategies like diversification and systematic investing. Addressing these biases is essential for fostering more rational and stable financial markets.

Ultimately, this study reinforces the need to incorporate behavioural insights into both financial models and market practices, enabling more effective regulation, improved decision-making, and greater market stability.

References

1. Barber, B. M., & Odean, T. (2000). Trading is Hazardous to Your Wealth: The Common Stock Investment Performance of Individual Investors. *SSRN Electronic Journal* , 2.
<https://doi.org/10.2139/ssrn.219228>
2. Berwal, R. (2017). Financial Inclusion in India: Role of RBI. *International Journal of Scientific and Research Publications*, 7(7). <https://www.ijsrp.org/research-paper-0717/ijsrp-p6755.pdf>
3. Bikhchandani, S., Hirshleifer, D., & Welch, I. (1992). A Theory of Fads, Fashion, Custom, and Cultural Change as Informational Cascades. *Journal of Political Economy*, 100(5), 992–1026.
4. Bikhchandani, S., & Sharma, S. (2001). Herd Behavior in Financial Markets: A Review. *SSRN Electronic Journal*, 47(3). <https://doi.org/10.2139/ssrn.228343>
5. de Mel, S., McKenzie, D., & Woodruff, C. (2008). Returns to Capital in Microenterprises: Evidence from a Field Experiment*. *Quarterly Journal of Economics*, 123(4), 1329–1372.
<https://doi.org/10.1162/qjec.2008.123.4.1329>
6. Diether, M., Scherbina, Chen, H., & Stein ; Du 41e, G. (1969). DotCom Mania: The Rise and Fall of Internet Stock Prices. *THE JOURNAL of FINANCE*, LVIII(3).
https://pages.stern.nyu.edu/~eofek/DotComMania_JF_Final.pdf
7. Fama, E. (1970). Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2), 383–417. <https://doi.org/10.2307/2325486>
8. JEGADEESH, N., & TITMAN, S. (1993). Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency. *The Journal of Finance*, 48(1), 65–91.
<https://doi.org/10.1111/j.1540-6261.1993.tb04702.x>

9. Kahneman, D., & Tversky, A. (1979). *Prospect Theory: an Analysis of Decision under Risk*.
https://web.mit.edu/curhan/www/docs/Articles/15341_Readings/Behavioral_Decision_Theory/Kahneman_Tversky_1979_Prospect_theory.pdf
10. Madrian, B. C., & Shea, D. F. (2001). The Power of Suggestion: Inertia in 401(k) Participation and Savings Behavior. *The Quarterly Journal of Economics*, 116(4), 1149–1187.
<https://doi.org/10.1162/003355301753265543>
11. Paúl, H., & Natarajan, S. (2023). Behavioral Biases and their Influence on Investment Decision-Making: A Systematic Literature Review and Future Research Agenda. *JOURNAL of LAW and SUSTAINABLE DEVELOPMENT*, 11(4), e904–e904. <https://doi.org/10.55908/sdgs.v11i4.904>
12. Thaler, R., & Sunstein, C. (2008). *Nudge: Improving decisions about health, wealth, and happiness*. Psycnet.apa.org. <https://psycnet.apa.org/record/2008-03730-000>
13. Gompers, P., & Lerner, J. (2001). The Venture Capital Revolution. *The Journal of Economic Perspectives*, 15(2), 145–168. <https://www.jstor.org/stable/2696596>